

# 第1頁 PAGE 1 — 能量攤位 / THE ENERGY STALL

花瓣仔的食物成分市場 / Petalino and the Food Substance Market



碳水化合物 → 葡萄糖 → 呼吸作用燃料 / Carbohydrates → glucose → respiratory fuel

**1** 葡萄糖可以裝成較大的儲存形式。  
Glucose can be packed into larger storage forms.

**2** 肝糖是動物的儲存形式，  
存於肝臟和肌肉。  
Glycogen is the storage form in animals, in the liver and muscles.

**3** 澱粉是植物的儲存形式，  
例如馬鈴薯。  
Starch is the storage form in plants, for example potato.

**4** 纖維素構成植物細胞壁。  
Cellulose makes the plant cell wall.

**5** 纖維素也是膳食纖維，  
可促進腸蠕動。  
Cellulose is also dietary fibre and promotes gut peristalsis.

**6** 動物存肝糖，植物存澱粉，  
纖維素作細胞壁和膳食纖維！  
Animals store glycogen, plants store starch, cellulose is cell wall and dietary fibre!



**1**

脂質小舖  
THE LIPID STALL

脂質可以儲存能量！  
Lipids can store energy!

能量儲存  
ENERGY STORAGE

**2**

脂質可以保暖，也可以保護器官。  
Lipids provide insulation and protect organs.

保暖  
INSULATION

器官保護  
ORGAN PROTECTION

**3**

脂質也能構成細胞膜。  
Lipids also form cell membranes.

細胞膜  
CELL MEMBRANE

**4**

一分子甘油加三分子脂肪酸。  
One glycerol plus three fatty acids.

甘油  
GLYCEROL

脂肪酸  
FATTY ACIDS

**5**

就會結合成三酸甘油酯，  
也就是脂肪和油！  
They join to make a triglyceride, which is fats and oils!

三酸甘油酯  
TRIGLYCERIDE

**6**

脂質：儲能、保暖、保護器官、  
構成細胞膜！  
Lipids: store energy, insulate, protect organs, form membranes!

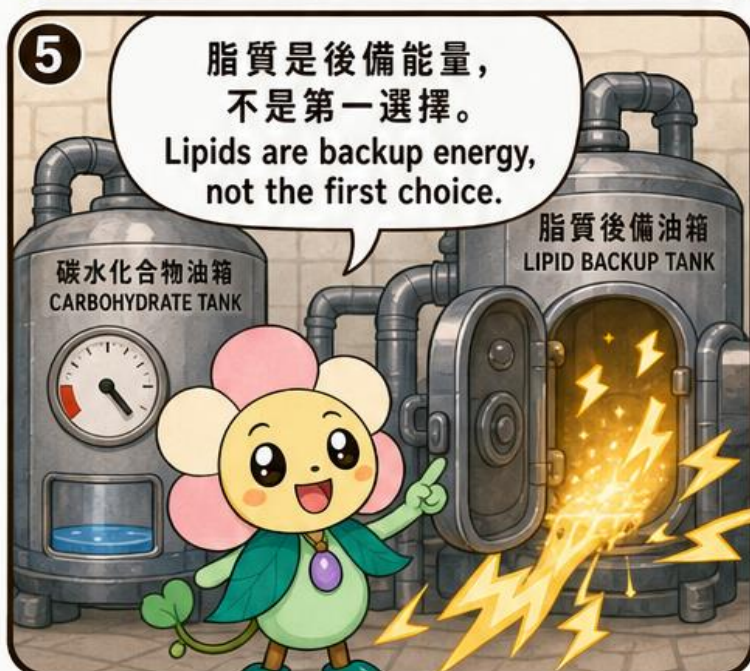
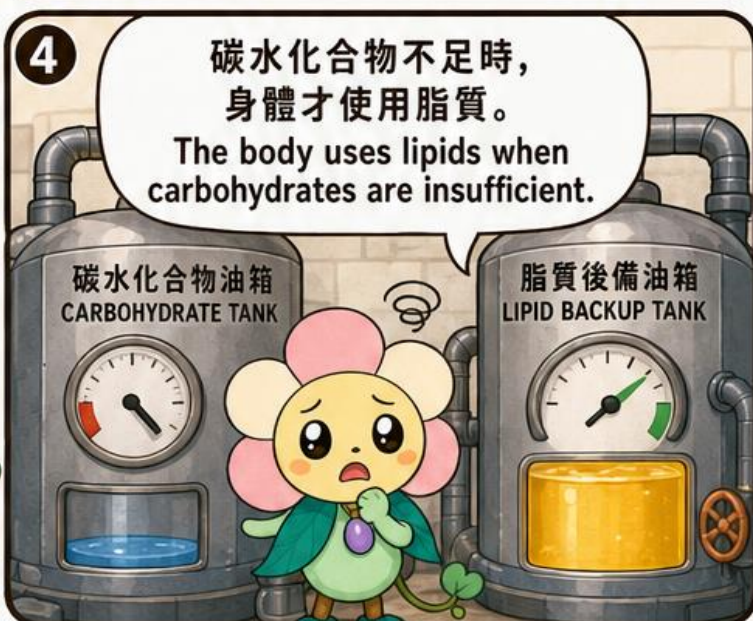
儲能  
STORE ENERGY

保暖  
INSULATE

保護器官  
PROTECT ORGANS

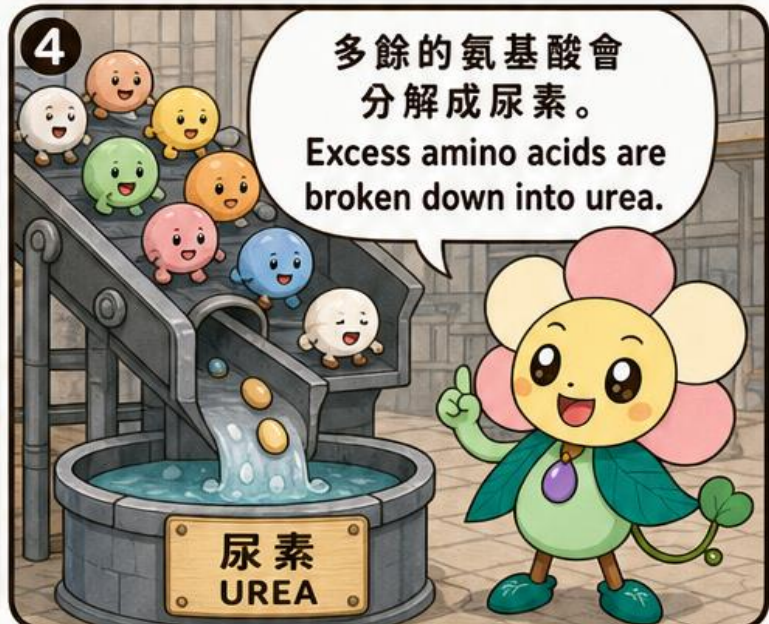
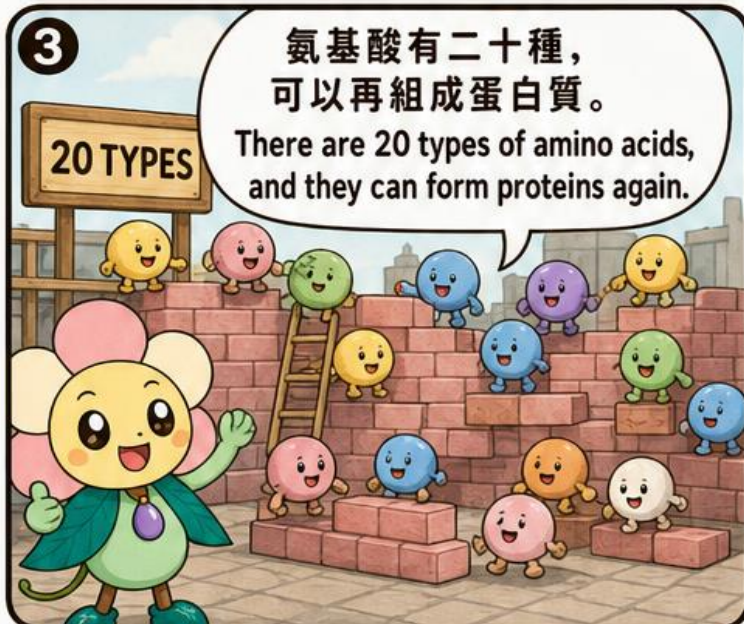
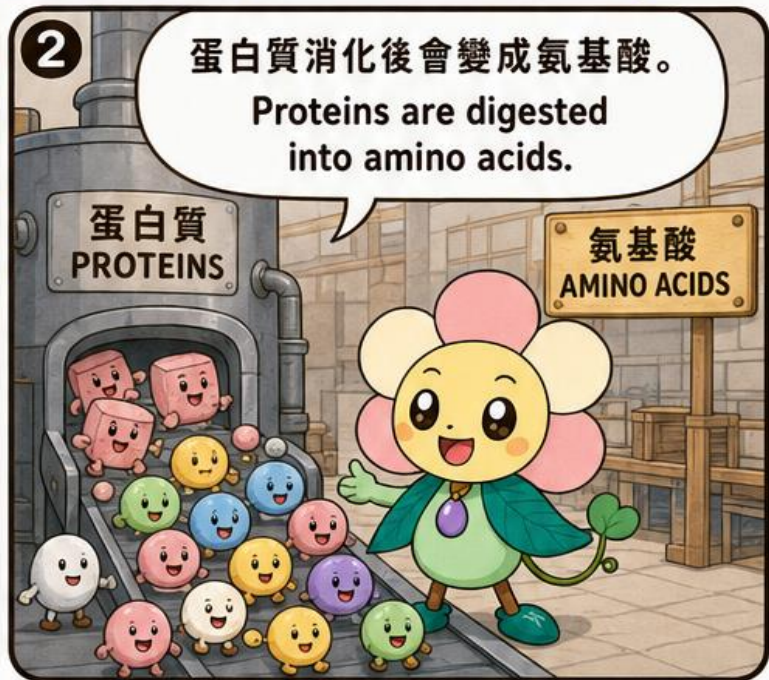
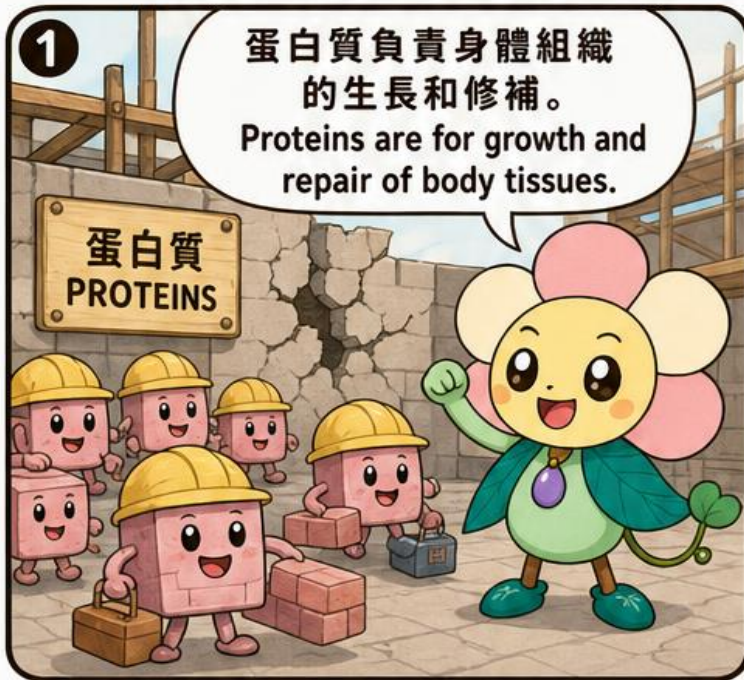
構成細胞膜  
FORM MEMBRANES

1 甘油 + 3 脂肪酸 = 三酸甘油酯 /  
1 glycerol + 3 fatty acids = triglyceride



碳水不足 → 才使用脂質 / Carbohydrates insufficient → then use lipids





蛋白質 → 氨基酸 → 生長修補 / 多餘變尿素 /  
Proteins → amino acids → growth and repair / excess becomes urea



維他命A → 昏暗視力 + 免疫力 / 缺乏 = 夜盲症 /  
Vitamin A → dim-light vision + immunity / deficiency = night blindness



**1** 維他命C保持皮膚和牙齦健康，也幫助傷口癒合。  
Vitamin C keeps skin and gums healthy and helps wounds heal.

**維他命C  
VITAMIN C**

**2** 缺乏會導致壞血病：牙齦出血，傷口癒合不良。  
Deficiency causes scurvy: bleeding gums and poor wound healing.

**壞血病  
SCURVY**

**3** 來源：新鮮水果和蔬菜。  
Sources: fresh fruits and vegetables.

**4** 維他命D保持骨骼和牙齒強壯。皮膚在陽光下也能製造它。  
Vitamin D keeps bones and teeth strong. The skin can also make it under sunlight.

**維他命D  
VITAMIN D**

**5** 缺乏會導致佝僂病：骨骼變軟和彎曲。  
Deficiency causes rickets: bones become soft and bent.

**佝僂病  
RICKETS**

**6** 來源：牛奶、魚肝油和蛋黃。也要曬適量陽光！  
Sources: milk, fish liver oil and egg yolk. Get some sunlight too!

**1** 礦物質工具箱有鈣、鐵和碘。  
The mineral toolbox has calcium, iron and iodine.

鈣  
CALCIUM

鐵  
IRON

碘  
IODINE

**2** 鈣構建骨骼和牙齒，也幫助血液凝固。  
Calcium builds bones and teeth and helps blood clotting.

鈣  
CALCIUM

**3** 兒童缺乏會佝僂病；成人缺乏會骨質疏鬆。  
In children, deficiency causes rickets; in adults, osteoporosis.

骨質疏鬆症  
OSTEOPOROSIS

乳製品  
DAIRY

深綠色蔬菜  
DARK GREENS

大豆  
SOYBEANS

**4** 鐵製造血紅蛋白。缺乏會貧血：蒼白、疲勞、暈倒。  
Iron makes haemoglobin. Deficiency causes anaemia: pale skin, tiredness and fainting.

貧血  
ANAEMIA

紅肉  
RED MEAT

肝臟  
LIVER

堅果  
NUTS

豆類  
BEANS

**5** 碘維持甲狀腺功能。缺乏會甲狀腺腫。  
Iodine maintains thyroid function. Deficiency causes goitre.

甲狀腺腫  
GOITRE

海鮮  
SEAFOOD

海藻  
SEAWEED

**6** 鈣建骨骼，鐵造血，碘護甲狀腺！  
Calcium builds bones, iron makes blood, iodine protects the thyroid!

鈣  
CALCIUM

鐵  
IRON

碘  
IODINE

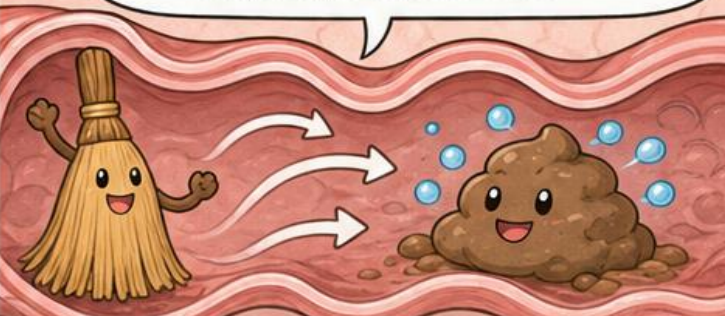
鈣 / 鐵 / 碘 → 佝僂病或骨質疏鬆、貧血、甲狀腺腫  
Ca / Fe / I → rickets or osteoporosis, anaemia, goitre

**1** 膳食纖維增加食物的體積。  
Dietary fibre adds bulk to food.



膳食纖維  
DIETARY FIBRE

**2** 它刺激腸道蠕動，並保持糞便柔軟。  
It stimulates peristalsis and keeps faeces soft.



腸道蠕動  
PERISTALSIS

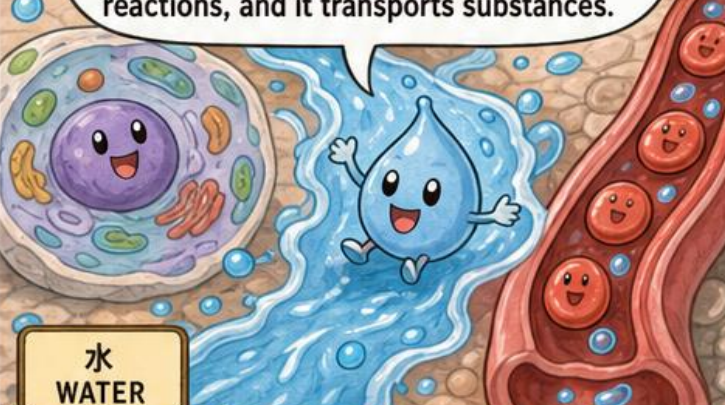
柔軟糞便  
SOFT FAECES

**3** 缺乏會便秘。  
來源：水果、蔬菜和全穀類。  
Deficiency causes constipation.  
Sources: fruits, vegetables and whole grains.



便秘  
CONSTIPATION

**4** 水是細胞質和血液的主要成分，也是化學反應的介質，並運輸物質。  
Water is the main component of cytoplasm and blood, a medium for reactions, and it transports substances.



水  
WATER

**5** 水溶解廢物、組成消化液，並靠汗液蒸發調節體溫。  
Water dissolves wastes, makes digestive juices, and regulates body temperature by sweat evaporation.



溶解廢物  
DISSOLVES WASTES

組成消化液  
MAKES DIGESTIVE JUICES

汗液蒸發降溫  
SWEAT EVAPORATION  
COOLS BODY

**6** 纖維和水都幫助排便。  
水還運輸、排廢和降溫！  
Fibre and water both help passing faeces. Water also transports, removes waste and cools the body!

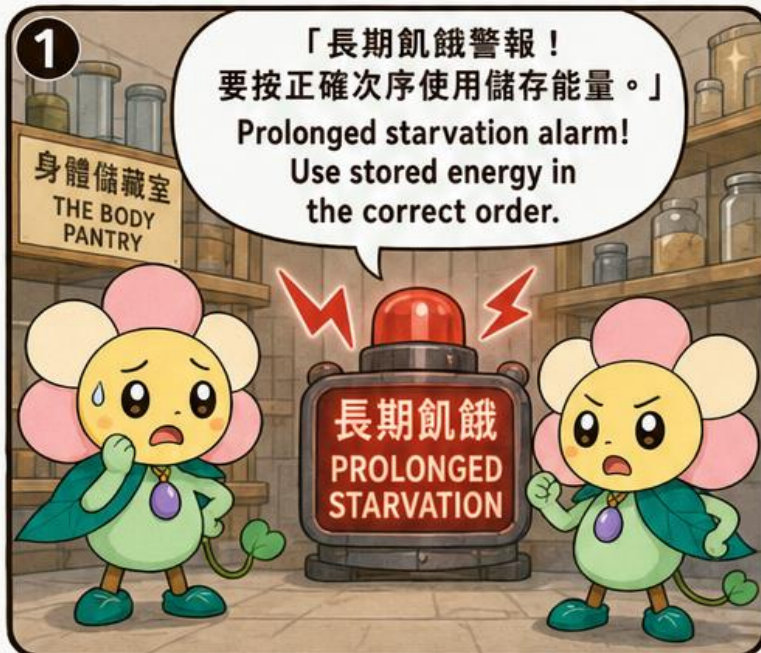


便秘  
CONSTIPATION



纖維 + 水 → 防便秘 / Fibre + water → prevent constipation





長期飢餓：碳水 → 脂質 → 蛋白質(最後) / 脂質能量 x2  
Starvation: carbs → lipids → proteins last / lipid energy x2